Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/24/2005
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Project Title:	Searching for the Best Protein Crystals: Integration of Synchrotron-Based Crystal Quality Measurements and Structure Determination		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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The following specific hypotheses have been developed to test the general hypothesis.
1. Long range order can be measured by mosaicity to give a physical indication of domain structure within the crystal (tested in aim 1).

2. CCD detectors with high spatial resolution can be used for rapid, quantitative topography (tested in aim 2).

3. Topography is also a measure of long range order and can be correlated with mosaicity measurements (tested in aim 3).

4. Diffuse scatter and resolution enhancement can be used as a measure of the crystalline short range order (tested in

Task Description:

aim 4).

5. Combining the data from aims 1-4 will produce a description of the crystal that can be used to understand basic physical processes during microgravity crystal growth, during cryocooling and for structure determination (tested in aim 5).

The following specific aims will be performed.

1. Software enhancement.

2. Develop CCD methods of quantitative topography.

3. Correlate topography to mosaicity and intensity.

4. Measurement of diffuse scatter and diffraction resolution.

5. Use these methods for structure determination and to explore the mechanisms of microgravity crystal growth and cryocrystallography. The general hypothesis of this research is as follows. Microgravity growth improves the long and short range order in macromolecular crystals. How this happens is not well understood. A complete physical characterization of macromolecular crystals will improve our understanding and enable us to obtain the maximum benefit from microgravity crystal growth, from cryocrystallography and also to collect the highest resolution diffraction data for structure determination. A recent review of biotechnology research on the international space station (Space Studies Board, National Research Council, Washington, DC 2000) recommended that "...NASA must focus its support on programs that are developing technologically innovative equipment and engaging in the structure determination of crystals with important biological implications". We are engaged in both of these areas and have a unique, interdisciplinary team with expertise in each area of the work.

We are working to produce tools that will provide a complete X-ray based physical characterization of a crystal in a short period of time. In combination with this we will determine high resolution structures. We will be able to relate the physical nature of the crystals to the resulting structural results and thereby evaluate the apparatus, mission or technique producing the crystal. We will gain an understanding of how to make optimal use of the valuable and limited microgravity environment. We can answer questions that until now have eluded researchers such as what is the best growth technique, and what acceleration disturbances can crystals tolerate and still be better than ground-grown crystals. The tools are not only applicable to microgravity systems. We can look at techniques used in everyday structural crystallography – e.g. what is the best method of cryocooling? How does annealing work, when it does improve diffraction? By applying our instrumentation to crystals including MnSOD, human replication protein A, and human serum albumin we gain not only knowledge about the physical process but structural detail to further the understanding of life itself.

Specifically we will study crystals of several important macromolecules. We will use the techniques described here to identify the best crystals and then to achieve the most detailed structure possible. Aim 1, improvements to existing software, will extend our experiments to provide even more physical information from the data collected. We will extract precise structural information from our superfine fsliced data. Topography (aim 2 and 3) will relate the components of mosaicity to the crystal and produce a visual image of the components contributing to the Bragg reflections. Quantitative measurements of the contributions from the components will allow accurate modeling of the crystal and contribute to highly precise intensity determination. Diffuse scatter (aim 4) will allow us to look at differences in long and short range order improvement. Achieving these aims will give us the most advanced capability for assessing a crystal. We will use this in aim 5 to ask the questions described above. Practically, how applicable is this? Very. Modern third generation synchrotrons are increasingly using unfocused radiation. We have commonly seen reflection profile structure that most integration software would not evaluate accurately. Our software developments will overcome this and will further structural determination in general. The tools developed will become available to all microgravity scientists for application to their specific research and is therefore of value to the microgravity community as a whole.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The primary aim of this research project is to identify the optimum methods for protein crystal growth by quantitatively evaluating crystal quality and to collect high-resolution X-ray diffraction data for structure determination. This involves growing crystals of several proteins, developing methods to quantitatively assess crystal quality and subsequently collecting high quality diffraction data for subsequent structure determination at a synchrotron. Crystal quality is assessed by diffraction resolution, topography and mosaicity. These parameters are measured simultaneously using an innovative technique combining unfocused synchrotron X-ray beam, a CCD area detectors and superfine phi slicing. Research in FY04 was focused on the continued development of a CCD camera for topography measurements. Substantial progress was made and includes:

• A CCD-based topography camera and corresponding software was developed in 2003 for the collection of X-ray topographs for the assessment of crystal quality. Initial, proof-of-concept, data were collected from insulin crystals at NSLS in February, 2003. A patent application was filed through UNMC on 10/2003. A manuscript on these early results was published in the J. Appl. Cryst. in 2004. This research was presented at a national meeting.

• Subsequently more CCD-based topography data was collected during 2004 with two synchrotron runs (in December ’03 and August ‘04) at the BioCARS beamline at APS. A bislide apparatus to hold the topography camera and position it in the correct location to collect topographs was tested. Software to coordinate data collection with the beamline hardware was tested. Data from room temperature and cryocooled lysozyme crystals was collected and published in J. Appl. Cryst. in 2004. In the second synchrotron run, these tools were used to compare room temperature and cryocooled crystals, this work will be presented at the 2005 American crystallography Association meeting in Orlando Florida and is in preparation for publication. A manuscript concerning “Ripple”, the topography control software, is also in preparation.

• A cryocooling/reversibility study on crystal quality was performed at the IMCA-CAT beamline at APS in collaboration with Drs. Doug Juers and Brian Matthews of the University of Oregon. Thousands of reflection profiles from thermally cycled beta galactosidase crystals were collected. Dr. Juers spent a week in Omaha in December, 2002 processing and

Task Progress:	analyzing the data and we have gone through multiple rounds of revision and this manuscript is nearly ready for submission. • For the structure determination of the semicrystalline state of profilin:actin project, the initial reflection profile data collection was published in J. Appl. Cryst., in 2004. The profilin:actin crystallization methodology has been taught to the Borgstahl laboratory by Dr. Narayan in person and the crystals have be subsequently reproduced by Dr. Borgstahl's technicians.. Dr. Schutt's precession camera has been moved from Princeton and installed in the laboratory in Omaha, NE. Most recently Mr. Edward Irobi joined the laboratory, before he moved from Uppsala University, Sweden to UNMC, he spent one month in Dr. Lindbergs laboratory in Stockholm learning to purify profilin:actin from calf thymus. Diffraction measurements on the semicrystalline state with the aim of solving this modulated structure have continued at UNMC. Summary of FY04 publications Published manuscript titles, citations and abstracts follow: 1. Vahedi-Faridi, A.; Lovelace, J.; Bellamy, H. D.; Snell, E. H.; and Borgstahl, G. E. O. "Physical and structural studies on the cryocooling of insulin crystals" Acta Cryst. D59, 2168-2182 (2003). Summary. Reflection profiles were analyzed from microgravity- (microg) and earth-grown insulin crystals to measure mosaicity (eta) and to reveal mosaic domain structure and composition. The effects of cryocooling on single and multi-domain crystals were compared. The effects of cryocooling on insulin structure were also re-examined. Microgravity crystals were of larger volume, more homogeneous, and of higher quality than earth crystals. Several microg crystals contained a single mosaic domain which encompasses all or nearly all of the crystal with etaavg of 0.005°. The earth crystals varied in quality and all contained multiple domains with etaavg of 0.031°. Cryocooling caused a 43-fold increase in eta for microg crystals (etaavg=0.217°) and an 8-fold increase for earth crystals (etaavg=0.246°). These results indicate that very well-ordered crystals are not completely protected from the stresses associated with cryocooling, especially when structural perturbations occur. However, there were differences in the reflection profiles. For multi-mosaic domain crystals, each domain individually broadened and separated from the other domains upon cryocooling. Cryocooling did not cause an increase in the number of domains. A crystal composed of a single domain retained this domain structure and the reflection profiles simply broadened. Therefore, an improved signal-to-noise ratio for each reflection was measured from cryocooled single domain crystals relative to cryocooled multi-domain crystals. The improved signal from microg crystals, along with the increase in crystal size, facilitated the measurement of the weaker high-resolution reflections. The observed broadening of reflection profiles indicates increased variation in unit cell dimensions which may be linked to cryocooling-associated structural changes and disorder. 2. Lovelace, J. J.; Soares, A. S.; Bellamy, H. D.; Sweet, R. M.; Snell, E. H.; and Borgstahl, G. E. O. "First results of digital topography applied to macromolecular crystals" J. Appl. Cryst. 37, 481-485 (2004). Summary. An inexpensive digital CCD camera was used to record x-ray topographs directly from large imperfect crystals of cubic insulin. Six reflections were recorded using a set of finely spaced stills encompassing the rocking curve of each reflection. Interesting and complex internal structure was observed by this technique. The CCD chip used in the camera has anti-blooming circuitry and produced good data quality even when pixels became overloaded. 3. Lovelace, J. J.; Narayan; K. Chik, J. K.; Bellamy, H. D.; Snell, E. H.; Lindberg, U.; Schutt C. E. and Borgstahl, G. E. O. "Three dimensional imaging of reflections from a modulated semi-crystalline state of profilin:actin" J. Appl. Cryst. 37, 327-330 (2004). Summary. Modulated protein crystals remain terra incognita for most crystallographers. While small molecule crystallographers have successfully wrestled with and conquered this type of structure determination, to date no modulated macromolecular structures have been reported. Profilin:actin in a modulated semi-crystalline state presents a challenge of sufficient biological significance to motivate the development of methods to accurately collect data on the complex diffraction pattern and to ultimately solve its structure. In this work, fine phi sliced data collection was used to resolve the closely-spaced satellite reflections from these polymorphic crystals. Image processing methods were used to visualize these data for comparison to original precession data. These preliminary data demonstrate the feasibility of using fine phi slicing to accurately collect the intensity and position of the main and satellite reflections from these modulated protein crystals. Manuscripts In Preparation FY04 . 1. Jeffrey J. Lovelace, Cameron R. Murphy, Henry D. Bellamy, KeithBrister, ReinhardPahl and Gloria E. O. Borgstahl "Advances in Digital Topography for Characterising Imperfections in Protein Crystals" submitted and revised for J. Appl. Cryst. (2004). 2. Gloria E. O. Borgstahl "How to Use Dynamic Light Scattering to Improve Your Chances of Growing Crystals of a Macromolecule" invited manuscript for Methods in Molecular Biology submitted in (2004). 3. Jeffrey J. Lovelace, Cameron R. Murphy, Keith Brister, Reinhard Pahl and Gloria E. O. Borgstahl, "Tracking reflections through cryogenic cooling with topography" in preparation (2004). 4. Juers, D. H., Lovelace, J., Bellamy, H. D., Snell, E. H., Matthews, B. W. and Borgstahl, G. E. O. "Reversibility in cryocooling crystal of e. coli beta galactosidase in preparation. 5. Vahedi-Faridi, Borgstahl, G. E. O. "Structure of the MnSOD-peroxide complex" in preparation.
Bibliography Type:	Description: (Last Updated: 08/10/2023)
Articles in Peer-reviewed Journals	Vahedi-Faridi A, Lovelace J, Bellamy HD, Snell EH, Borgstahl GEO. "Physical and structural studies on the cryocooling of insulin crystals." Acta Crystallographica Section D. 2003 Dec;59(12):2169-82. http://dx.doi.org/10.1107/S0907444903019668 , Dec-2003

Articles in Peer-reviewed Journals	Lovelace JJ, Soares AS, Bellamy HD, Snell EH, Borgstahl GEO. "First results of digital topography applied to macromolecular crystals." Journal of Applied Crystallography. 2004 Jun;37(3):481-5. http://dx.doi.org/10.1107/S0021889804004303 , Jun-2004
Articles in Peer-reviewed Journals	Lovelace JJ, Narayan K, Chik JK, Bellamy HD, Snell EH, Lindberg U, Schutt CE, Borgstahl GEO. "Imaging modulated reflections from a semi-crystalline state of profilin:actin crystals." Journal of Applied Crystallography. 2004 Apr;37(2):327-30. http://dx.doi.org/10.1107/S0021889804001773 , Apr-2004
Awards	Borgstahl G. "Selected as member of American Crystallography Association Continnuing Education Committee, January 2004." Jan-2004
Presentation	Borgstahl, G. E. O. "Towards the Structure Determination of a Modulated Protein Crystal: A Semicrystalline State of Profilin:Actin" APS Users Science Seminar Dec-2003
Presentation	Borgstahl, G. E. O. "Towards the Structure Determination of a Modulated Protein Crystal: A Semicrystalline State of Profilin:Actin" Department of Chemistry, University of Kentucky Oct-2003
Presentation	Borgstahl, G. E. O.; Murphy, C.; Lovelace, J. J.; Narayan, K.; Svensson, C.; Lindberg U.; and Schutt, C. E. "Towards the Structure Determination of a Modulated Protein Crystal: A Semicrystalline State of Profilin:Actin" FEBS meeting on Cytoskeletal Dynamics, Helsinki, Finland Jun-2004